Numerical Solution of the 1-D Time-Independent Schrödinger Equation

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Abstract

The time-independent one-dimensional Schrödinger equation was solved numerically in dimensionless form for a particle confined to a box, $x \in [0, L]$, for both a flat (infinite square well) and a harmonic potential. After discretizing and non-dimensionalizing the equation, energy eigenvalues were found with a shooting method, which halves and reverses the energy increment each time the wave function at the $x = L$ boundary changes sign. For the infinite well, the eigenvalues agreed with the analytic values up to the tolerance of the method (chosen to be 10^{-6}, though values down to $\sim 10^{-12}$ could be used, where floating-point rounding errors cap precision). For the harmonic potential, the error grew with the eigenstate index, since higher states have non-negligible tails at the boundaries. All computed eigenstates were checked to satisfy the Heisenberg uncertainty relation. The spacing between consecutive eigenvalues was constant for the lowest harmonic states and linearly increasing with n for higher harmonic states, matching the infinite square well. This is because higher energies made the harmonic oscillation between the boundaries negligible.

1 Introduction

The following lab report discusses a numerical approach to solving the 1-D time-independent Schrödinger equation and finding dimensionless solutions for energy eigenstates.

1.1 Schrödinger Equation

We take as our starting point the general 1-D time-independent form of the Schrödinger equation [1]:

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x). \tag{1.1}$$

For simplicity we consider the particle of mass m to be bound by infinite wells at $x = 0$ and $x = L$. We can then non-dimensionalize the equation to avoid working with small numbers such as $\hbar$ and arrive at the equivalent form of

$$\frac{d^2\psi(\tilde{x})}{d\tilde{x}^2} + \gamma^2\left(\epsilon - \nu(\tilde{x})\right)\psi(\tilde{x}) = 0 \tag{1.2}$$

with $\nu(\tilde{x}) = \frac{V(\tilde{x})}{V_0}$, $\epsilon = \frac{E}{V_0}$, $\gamma^2 = \frac{2mL^2V_0}{\hbar^2}$, and $\tilde{x} = \frac{x}{L}$ [1]. For this lab, we first set $\nu(\tilde{x}) = -1$ for the infinite square well, and then compare with the harmonic potential $\nu(\tilde{x}) = 8(\tilde{x} - 0.5)^2 - 1$.

1.2 Numerical Discretization of the Wave Function

The previous non-dimensionalization (1.2) allows the Schrödinger equation to be discretized in order to study it numerically. By splitting up $\tilde{x}$ into N intervals of length $l = \frac{1}{N-1}$, we can define a discrete wave function with

$$\psi_n(\tilde{x}) = \psi(\tilde{x}_n) = \psi(nl) \tag{1.3}$$

where n takes values from 0 to $N-1$, inclusive [1]. Note that $\tilde{x}$ ranging from 0 to 1 is henceforth referred to as x for simplicity, as the original x ranging from 0 to L is not used. With this discrete form of the wave function, we can find a recursive definition such as

$$\psi_{n+1} = \frac{2\left(1 - \frac{5}{12}l^2k_n^2\right)\psi_n - \left(1 + \frac{1}{12}l^2k_{n-1}^2\right)\psi_{n-1}}{1 + \frac{1}{12}l^2k_{n+1}^2} \tag{1.4}$$

where $k_n^2 = \gamma^2(\epsilon - \nu(x_n))$ [1]. This value (k_n^2) varies depending on which potential is considered, not just because of the explicit $\nu(x)$ dependence, but also because I chose $\gamma^2 = 200$ for the infinite well potential and $\gamma^2 = 1000$ for the harmonic potential. A larger γ^2 narrows the harmonic potential's wave functions, which reduces the numerical error. The reason for this is discussed after the results shown in Table 1. Note that the recursive relation defined in (1.4) requires two points to find a third. Though we define $\psi_0 = 0$, we must also arbitrarily set ψ_1 to be able to generate the wave function (I chose $\psi_1 = 10^{-4}$). This value scales the wave function, but this degree of freedom was removed when the wave function was normalized, so my choice did not affect any results.

```
def psi_eps(eps, x=x):
  k2 = [gamma2 * (eps + 1) for i in x]
  psi = [psi_0, psi_1]
  for i in range(1, N - 1):
    psi.append((2 * (1 - 5/12 * l**2 * k2[i]) * psi[i] -
     (1 + 1/12 * l**2 * k2[i-1]) * psi[i-1]) / (1 + 1/12
     * l**2 * k2[i+1]))
  return psi
```

The code above is equivalent to (1.4) for $\nu(x) = -1$, where `psi_0` $= 0$, `psi_1`, `l`, `gamma2`, `N`, and `x` are defined elsewhere. The function returns a list, `psi`, containing the discretized wave function.

2 Methods

2.1 Finding the Energy Eigenstates: The Shooting Method

An energy eigenstate (found for a discrete wave function) fulfills the boundary condition $\psi(N-1) = 0$. The other boundary condition, that the wave function vanishes at the origin, was set manually. The method used to find an eigenstate that fulfills both conditions is called the shooting method.

Starting at a trial energy $\epsilon = \epsilon_0$, the corresponding wave function was evaluated at the boundary, $\psi_\epsilon(N-1)$. The energy was then increased by an increment $\Delta\epsilon$ and $\psi_{\epsilon+\Delta\epsilon}$ was reevaluated. At this point, the trial energy value was reset: $\tilde{\epsilon} = \epsilon + \Delta\epsilon$. As for the increment, if the sign of $\psi(N-1)$ differed for ϵ and $\tilde{\epsilon}$, $\Delta\epsilon$ was changed to $\tilde{\Delta\epsilon} = -\frac{\Delta\epsilon}{2}$. This makes the energy sequence ϵ_n converge around ϵ^*, the energy eigenstate, for $n \to \infty$. Numerically, since it is unlikely that $\psi_{\epsilon_n}(N-1) = 0$ is exactly reached, the iteration was stopped once the increment satisfied $|\Delta\epsilon| <$ `acc`, so that ϵ was determined to within `acc`. The code for this method is shown below:

```
def find_eps(eps_0, acc=10**-6):
  eps = eps_0
  delta_eps = 0.04
  while abs(delta_eps) >= acc:
    prev_eps = eps
    eps += delta_eps
    if psi_eps(prev_eps)[N-1] * psi_eps(eps)[N-1] < 0:
      delta_eps /= -2
  return eps
```

2.2 Normalizing the Wave Functions

As previously mentioned, scaling the wave function by a factor does not alter the boundary conditions. The condition we consider in order to normalize the wave function is that the probability of finding the particle anywhere along the wave function must be 1:

$$\int_0^1 |\psi(x)|^2 dx = 1. \tag{2.1}$$

With this, for any wave function $\psi(x)$ we can define a normalized wave function:

$$\tilde{\psi}(x) = \frac{\psi(x)}{\sqrt{\int_0^1 |\psi(x)|^2 dx}}. \tag{2.2}$$

For the discrete wave function this is quite straightforward.

```
def norm_psi(eps, x=x):
  factor = np.sqrt(simpson([i**2 for i in psi_eps(eps)], x=x))
  return [i / factor for i in psi_eps(eps)]
```

The normalization function takes the parameter eps, the eigenstate ϵ for the specific wave function, in order to generate the wave function as defined earlier in the function psi_eps. It then returns the normalized wave function, scaled by the factor as shown in (2.2). A standard check that the value of $\int_0^1 |\tilde{\psi}_n(x)|^2 dx$ is 1 shows that it is, in fact, normalized. The normalization can be seen below for two different forms of $\nu(x)$ representing the infinite well and harmonic potentials.

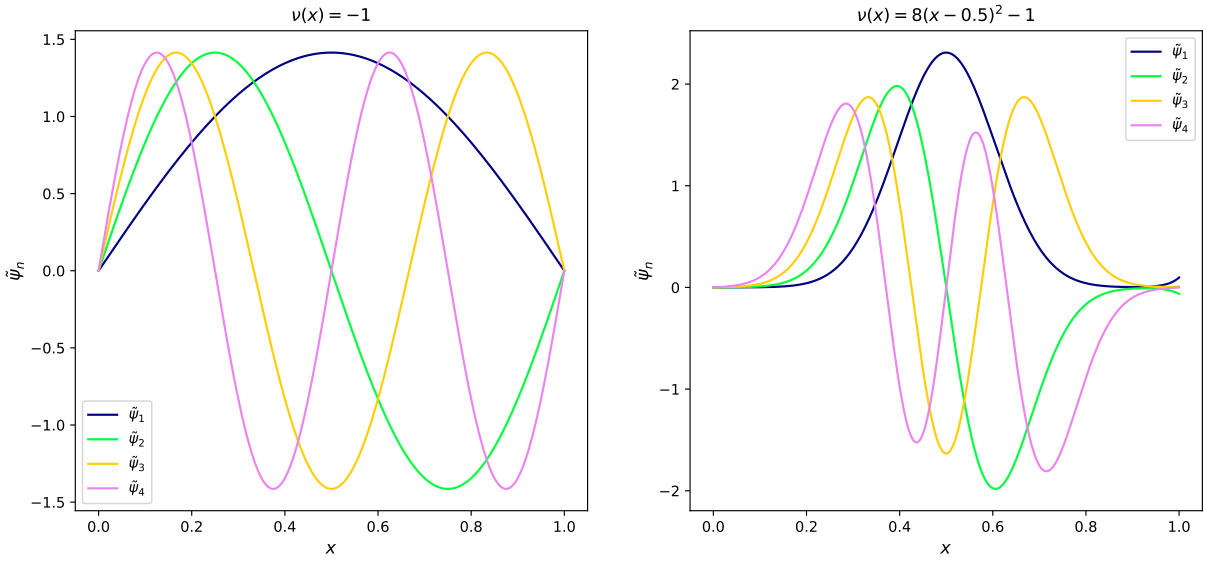


Figure 1: Plot of normalized wave functions for the first four energy eigenstates of the infinite well (left) and harmonic (right) potentials.

3 Results and Discussion

3.1 Energy Eigenvalues

Using the shooting method as detailed in Section 2.1 for both $\nu(x) = -1$ and $\nu(x) = 8(x-0.5)^2 - 1$ (the infinite well and harmonic potentials, respectively), the following numerical values were obtained. The analytic expected values are listed next to them, obtained from [2], sections 2.2 and 2.3.

	Infinite well ($\gamma^2 = 200$)		Harmonic ($\gamma^2 = 1000$)	
n	Numerical	Analytic	Numerical	Analytic
1	−0.950653	−0.950651	−0.9105575	−0.9105572
2	−0.802608	−0.802607	−0.731672	−0.731671
3	−0.555869	−0.555867	−0.5527868	−0.5527864
4	−0.2104318	−0.2104316	−0.37389	−0.37390
5	0.23369	0.23370	−0.19499	−0.19501
6	0.776527	0.776528	−0.0160	−0.0161
7	1.418052	1.418053	0.1634	0.1628
8	2.158272	2.158273	0.344	0.342
9	2.997188	2.997189	0.529	0.521
10	3.934801	3.934802	0.72	0.70

Table 1: Numerical and analytic eigenvalues for the infinite well and harmonic potentials, with decimal precision kept to the first discrepancy across numerical and analytic values.

The infinite well potential maintained a precision of $\sim 10^{-6}$ for all n, as the only error came from the tolerance of the shooting method. The harmonic potential, on the other hand, had an error that grew with n: higher-energy states extend further towards the walls, and forcing them to zero there shifts the numerical eigenvalues above the analytic ones. I chose $\gamma^2 = 1000$ for this potential because it narrows the wave functions, so their tails reach the walls only at higher n.

The error in the numerical method was also studied as a function of the parameter `acc` for the infinite well:

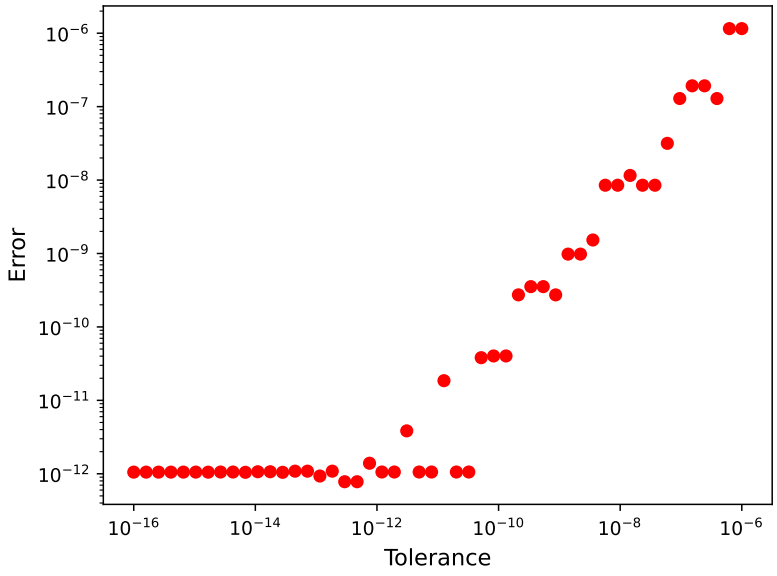


Figure 2: Plot of the error in the first energy eigenstate for different choices of `acc` in the `find_eps` function.

The above figure shows that increasing the tolerance `acc` generally increases the error in the first energy eigenstate, at least above `acc` $\approx 10^{-13}$. In this range, the error is similar to a step function. It depends entirely on at what value of $\Delta\epsilon$ the eigenstate is found. Since $|\Delta\epsilon_k| = \frac{|\Delta\epsilon_0|}{2^k}$, any tolerance `acc` $\in (\Delta\epsilon_0 2^{-k-1}, \Delta\epsilon_0 2^{-k}]$ should yield the same error. It is worth noting that some values may land very near the root earlier in the iteration by coincidence, causing a few dips in the error.

Moreover, in Figure 2, for tolerances under 10^{-13} the error is constant at $\epsilon_{min} \approx 10^{-12}$. The

reason is rounding errors: when values get small enough, smaller decimals are rounded away, so there is no way to obtain higher precision. This is the limiting factor in obtaining the highest possible precision for the eigenstates.

3.2 Heisenberg Uncertainty Relations

In the eigenstates (and in general), the wave function must obey the Heisenberg uncertainty principle, which in dimensionless coordinates is $\Delta x \Delta p \geq \frac{1}{2}$.

Using the usual standard deviation formula, $\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$, I calculated the uncertainty (or standard deviation) in x with the code below.

```
def stdev_x(psi, x=x):
    mean  = simpson(x * psi**2, x=x)
    mean2 = simpson(x**2 * psi**2, x=x)
    return np.sqrt(mean2 - mean**2)
```

The uncertainty in p was calculated with the formula from [3] with $\hbar \equiv 1$ and $\langle p \rangle = 0$:

$$\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{-\hbar^2 \int_0^1 \psi \psi'' dx} = \sqrt{-\int_0^1 \psi \psi'' dx}. \tag{3.1}$$

The second derivative of ψ was approximated with the central difference method (explained in [4]), which loses the first and last term of the interval. This does not change the final result, however, as $\psi(0) = \psi(N-1) = 0$, so the contribution to $\psi\psi''$ would be 0 anyway. The code equivalent of (3.1) is the following:

```
def stdev_p(psi, x=x):
    l = x[1] - x[0]
    d2psi = (psi[:-2] - 2*psi[1:-1] + psi[2:]) / l**2
    return np.sqrt(-simpson(psi[1:-1] * d2psi, x=x[1:-1]))
```

Using both standard deviations as uncertainties, the following plot was obtained for the uncertainty products of each eigenstate:

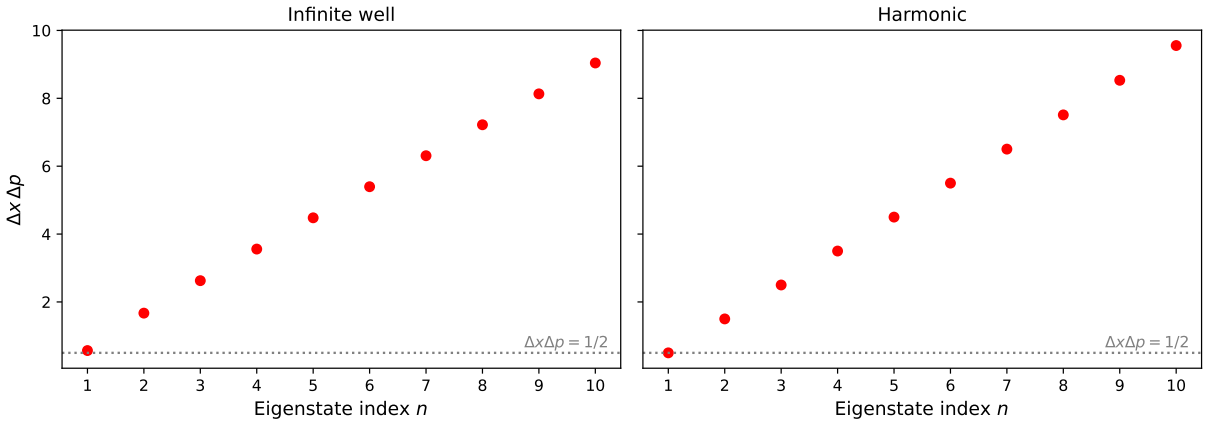


Figure 3: Plot showing the uncertainty products for each eigenstate for both potentials.

As is clear in Figure 3, all of the eigenstates fulfill the Heisenberg uncertainty principle for both potentials, both infinite well and harmonic.

4 Eigenstate Energy Differences

The energy increments between consecutive eigenvalues can also be checked for the harmonic potential:

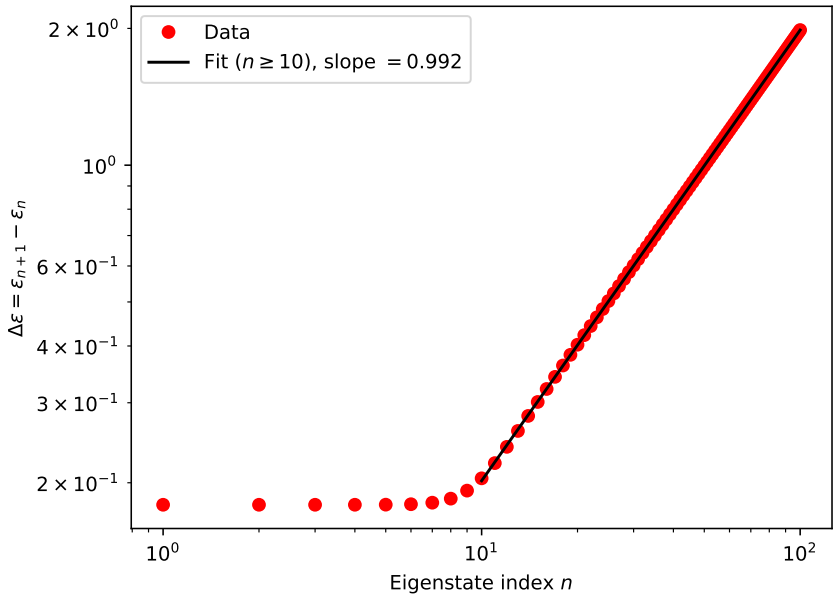


Figure 4: Plot of the difference between consecutive eigenvalues and the eigenstate index n.

When the energy difference is plotted against the index on a double logarithmic scale, it is basically flat, or constant, for the first few indices. This indicates equally spaced eigenvalues. However, for higher values of n, the slope changes. Eventually, the slope tends to 1, which indicates $\Delta\epsilon \propto n$. Physically, this represents the state in which the particle is so excited that the harmonic potential inside the bounds is negligible: it recreates the square well potential. I used a higher value of $n_{\max}$ than recommended in [1] as it made the slope $= 1$ clearer.

5 Conclusions

The shooting method reproduced the first 10 eigenvalues of the infinite well to the tolerance `acc`, chosen to be 10^{-6}. Additionally, the error between the numerical and analytic values (measured for the first eigenstate of the infinite well potential) followed a step-like dependence on `acc`. Arbitrarily high precision is impossible, nonetheless, due to machine rounding. This caps the precision at $\sim 10^{-12}$.

Furthermore, all computed eigenstates satisfied the dimensionless Heisenberg uncertainty relation $\Delta\tilde{x}\,\Delta\tilde{p} \geq 1/2$. The product increased with n and the ground state ($n = 1$) was just above the bound for both potentials.

Finally, the eigenvalue spacing $\Delta\epsilon$ for the harmonic potential was constant for the lowest states but crossed over to a $\Delta\epsilon \propto n$ regime near $n \approx 10$, approaching the same behavior as the infinite well.

This lab has shown that the time-independent 1-D Schrödinger equation can be discretized to find numerical solutions. The accuracy of these solutions depends on the methods used, but the results will generally be precise, especially for the first few eigenstates, i.e. for lower energies.

References

[1] Paul Eastham, Mark Mitchison, and Matthias E. Möbius. Numerical solution of the time-independent 1-d Schrödinger equation. Computational lab script (JS Computation Lab), 2023.

[2] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. Cambridge University Press, 3 edition, 2018.

[3] Marc R. Roussel. The particle in a box and the uncertainty principle. Lecture notes, Department of Chemistry and Biochemistry, University of Lethbridge.

[4] Ruye Wang. Derivative approximated by finite difference. Lecture notes, Harvey Mudd College. Accessed: 2026-10-26.